



फलसु



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Part I: Multiple Choice

Choose the best answer from the suggested options and write the letter of your choice

ONLY on the answer sheet (1 pt. each).

1) In Economics, fixed inputs are defined as those inputs

- A) Whose supply can be altered in response to desired changes in output
- B) Which have direct relationship with each units of output produced
- C) Which are used in a constant proportion with other inputs
- ☒ D) Whose quantity do not change over the short run despite a change in the demand of the final product

2) What should the consumer do to maximize his/her utility if $MU_A/P_A > MU_B/P_B$, where A and B are consumption goods and P_A and P_B are respective market prices?

- ☒ A) Consume more of good A
- B) Consume more of good B
- C) Consume more of good A and B
- D) Consume less of good A and B

$$\frac{MU_A}{P_A} > \frac{MU_B}{P_B} \Rightarrow MU_A = MU_B$$

3) Which one of the following is correct about stages of production?

- ☒ A) Stage I ranges from origin to the level of labor input where APL is maximum.
- B) Stage II ranges from a level of labor input where MPL is maximum to where MPL equals zero
- C) Stage III is above the level of labor input where APL is maximum
- D) Stage II is the stage of efficient production, while stage I and III are not.

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4) Budget line is

- A) Rotates clockwise when price of y increases and price of x remains constant
- ☒ B) Rotates clockwise when price of y decreases and price of x remains constant
- C) Rotates anti-clockwise when price of x increases and price of y remains constant
- D) Rotates clockwise when price of x decreases while price of y remains constant

5) Which one of the following is the limitation of national income accounting

- A) It is difficult to capture non-marketed productive activities such as domestic activities, informal business, and volunteer services.
- B) It does not take into account quality differences among goods and services.
- C) Natural resource and environmental damage is not accounted.
- ☒ D) All

6) One of the following is NOT valid about stages of short-run production?

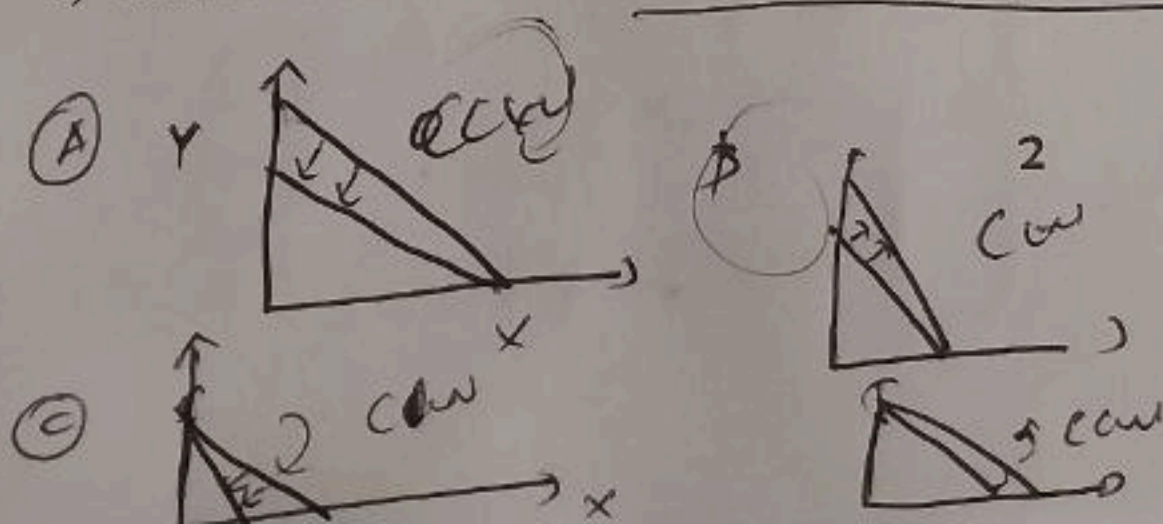
- A) In stage I marginal product of the variable input is higher than its average product
- B) Stage III is also known as stage of negative marginal return
- ☒ C) Firms will be maximizing profits only if they are operating in Stage II
- D) Stages I & II are customarily known as economic regions in production process

7) The market price in a perfectly competitive firm is (Birr 60). If a firm has a total fixed cost (TFC) of birr 10,000 and average variable cost (AVC) of birr 40, then the break even quantity will be:

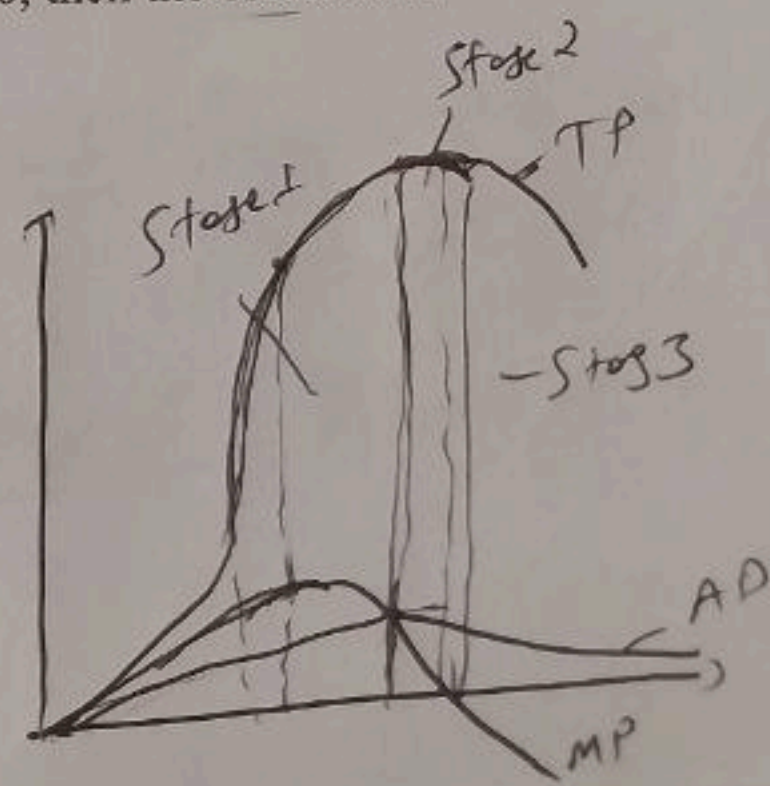
- A) 60,000
- B) 40
- C) 80
- ☒ D) 500

$$\begin{aligned} ATC &= P \\ \frac{10,000}{q} + 40 &= 60 \\ \frac{10,000}{q} &= 20 \\ q &= 500 \end{aligned}$$

8) Indifference curves are downward slopping to the right because



$$MP = AP$$



- D) In the long run, the average total costs of the firm will eventually diminish
- 29) Suppose you are going to be graduated from this university in the coming year and after being graduated, you are going to be considered as unemployed until you get job in certain industry. The type of unemployment you would probable experience is

☒ A) Frictional
☐ B) Cyclical
☐ C) Structural
☐ D) Natural

- B 30) If government expenditure exceeds tax collection, it is termed as:

☐ A) Budget surplus
☒ B) Budget deficit
☐ C) excess revenue
☐ D) trade deficit

- 31) A policy tool which is concerned with the use of tax and government expenditure is:

☐ A) Price and income policy
☐ B) International policy
☐ C) Monetary policy
☒ D) Fiscal policy

- 32 Two firms, Ethio telecom and Safaricom dominate the Ethiopian telecom industry.

Such type of market structure is known as:

☐ A) Monopolistically competitive
☐ B) Pure Monopoly
☒ C) Duopoly
☐ D) Free market

- 33) An individual consuming two commodities X and Y has the utility function $U(X, Y) = 2X^{0.5}Y^{0.5}$. The prices of the two commodities are 2 birr and 1 birr respectively. The consumer has a total income of 100 birr to be spent on the two goods. Then equilibrium levels of X, Y, and the slope of the budget line respectively are:

☒ A) 20, 60, 2
☐ B) 2, 60, 20
☐ C) 60, 20, 2
☐ D) None

- 34) The theory of consumer behavior assumes that:

☒ A) Consumers behave rationally, maximizing their satisfaction.
☐ B) Consumers have unlimited money incomes.
☐ C) Consumers do not know how much marginal utility they obtain from successive units of various products.
☐ D) Marginal utility is constant.

- 35) During the Covid-19 many firms shutdown the organization, particularly hotel sector and tourism sectors were stopped the operation and laid-off the workers. Generally, the national economy has fallen, unemployment has raised, operation has fallen, etc. during Covid-19. An unemployment because of these economic crises is called:

☐ A) Structural unemployment
☐ B) Frictional unemployment
☒ C) Cyclical unemployment
☐ D) Disguised unemployment

- 36) Measuring national income is important for:

☐ A) To compare the living standard of citizens among different countries.
☐ B) To understand the size of the economy.
☐ C) To formulate different economic policies.
☒ D) All of the above

- 37) _____ is a cost that a firm cannot avoid even by producing zero level of output in the short run

☐ A) Average variable cost
☐ B) Marginal cost
☒ C) Fixed Cost
☐ D) Variable cost

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$P = 100 - 2Q_d$
 $2Q_d = 100 - P$
 $Q_d = 50 - \frac{P}{2}$

$\frac{P_x}{P_y} = 2$

$\frac{MU_x}{MU_y} = \frac{P_x}{P_y} = 2$

$2X + Y = 100$
 $5X = 100$
 $X = 20$

$Y = 20 \times 3$
 $Y = 60$

$\frac{4Y^{\frac{2}{3}}}{5X^{\frac{2}{3}}} = \frac{6 \times \frac{2}{3}}{5 \times \frac{2}{3}}$

$4Y = 12X$
 $Y = 3X$

$MU_x = 2 \left(\frac{2}{3} \right) X^{-\frac{1}{3}} Y^{\frac{1}{2}}$

$MU_y = 2 \left(\frac{1}{2} \right) X^{\frac{1}{2}} Y^{-\frac{1}{2}}$

$\frac{4}{5} X^{-\frac{1}{3}} Y^{\frac{1}{2}} = \frac{4}{6} X^{\frac{1}{2}} Y^{-\frac{1}{2}}$
 $\frac{Y^{\frac{3}{2}}}{X^{\frac{2}{3}}} = \frac{5}{6} X^{\frac{5}{6}} Y^{-\frac{1}{6}}$
 $\frac{Y^{\frac{3}{2}}}{X^{\frac{2}{3}}} = \frac{5}{6} X^{\frac{5}{6}} Y^{-\frac{1}{6}}$

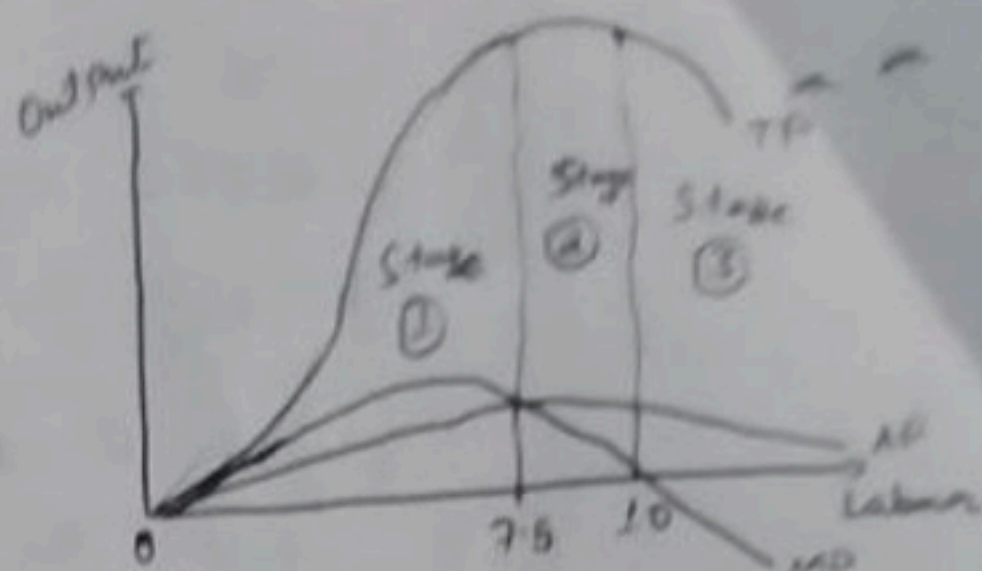
②

Given
K = 1

$$Q(K, L) = -2L^3 K^2 + 30L^2 K$$

Solution

$$Q(L, L) = -2L^3 + 30L^2$$



$$a) AP_L = \frac{Q}{L} = \frac{-2L^3 + 30L^2}{L} = -2L^2 + 30L \quad \text{--- (1)}$$

$$MP_L = \frac{\Delta Q}{\Delta L} = \frac{dQ}{dL} = d(-2L^3 + 30L^2)$$

$$MP_L = -6L^2 + 60L \quad \text{--- (2)}$$

$$b) AP_{L \max} = \frac{d(AP_L)}{dL} = \frac{d(-2L^2 + 30L)}{dL} = 0$$

$$-4L + 30 = 0 \Rightarrow 4L = 30$$

$$L = 7.5$$

$$Q(7.5) = -2L^3 + 30L^2 = -2(7.5)^3 + 30(7.5)^2$$

$$= -843.75 + 1687.5 = \underline{\underline{843.75}}$$

c) Stage I: From $L=0$ to $MP_L = AP_L$ point or $AP_{L \max}$

$$-6L^2 + 60L = -2L^2 + 30L$$

$$-6L^2 + 2L^2 = 30L - 60L$$

$$\Rightarrow -4L^2 = -30L$$

$$-4L^2 + 30L = 0 \Rightarrow L = 7.5$$

Stage II from $L=7.5$ to

$$MP_L = 0$$

$$-6L^2 + 60L = 0$$

$$-6(L^2 - 10L) = 0 \Rightarrow L(L-10) = 0$$

$$L \neq 0, \quad L = 10$$

Stage III: Above $L=10$

Stage I: from $L=0$
to $L=7.5$

Stage II: from $L=7.5$ to
 $L=10$

Stage 3: Above $L=10$

Multiple Choice

1. D ✓	11. A ✓	21. C ✓	31. D ✓
2. A ✓	12. D X	22. B ✓	32. C ✓
3. A ✓	13. A ✓	23. A ✓	33. A ✓
4. B ✓	14. A ✓	24. D ✓	34. A ✓
5. D ✓	15. C ✓	25. B ✓	35. C ✓
6. C X	16. A X	26. B ✓	36. D ✓
7. D ✓	17. B ✓	27. B ✓	37. C ✓
8. A ✓	18. A ✓	28. B ✓	38. C ✓
9. C ✓	19. C X	29. A ✓	39. D ✓
10. A ✓	20. D ✓	30. B ✓	40. D ✓

WORK OUT~~Question~~

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① Given :

Market Demand - - - $P = 100 - 2Q \Rightarrow Q_d = 50 - \frac{P}{2}$
 ~~$Q_d = 50 - \frac{P}{2}$~~

Market Supply - - - $P = 10 + Q \Rightarrow Q_s = P - 10$

$$TC = 20 + 40Q + 6Q^2 - Q^3$$

Solution② Market Equilibrium - - - $Q_d = Q_s$

$$50 - \frac{P}{2} = P - 10 \Rightarrow 60 = P + \frac{P}{2} \Rightarrow 60 = \frac{3P}{2}$$

$$60(2) = 3P \Rightarrow P = \frac{60(2)}{3} = \underline{40}$$

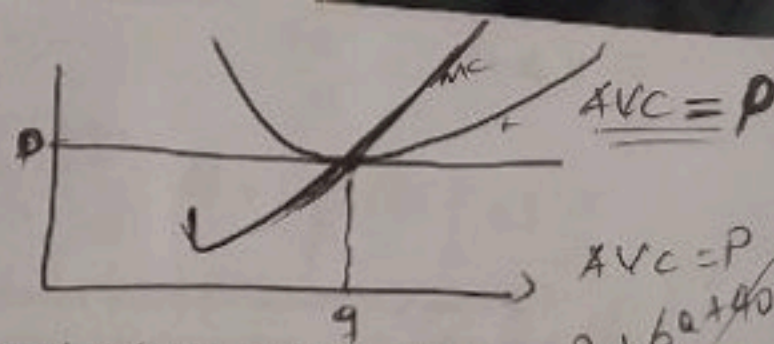
to maximize Profit:

$$\frac{d(\pi = TR - TC)}{dQ} \Rightarrow \frac{d\pi}{dQ} = \frac{d(TR)}{dQ} - \frac{d(TC)}{dQ} = 0$$

$$\Rightarrow 0 = MR - MC \Rightarrow \underline{MR = MC}$$

$$MR = \frac{\Delta TR}{\Delta Q} = \frac{d(PQ)}{dQ} = \frac{d(40Q)}{dQ} = \underline{40} \quad \text{--- ①}$$

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{d(TC)}{dQ} = \frac{d(-Q^3 + 6Q^2 + 40Q + 20)}{dQ} = -3Q^2 + 12Q + 40$$



$$AVC = P$$

$$-Q^2 + 6Q + 40 = 40$$

$$-Q^2 + 6Q = 0$$

$$Q^2 - 6Q = 0$$

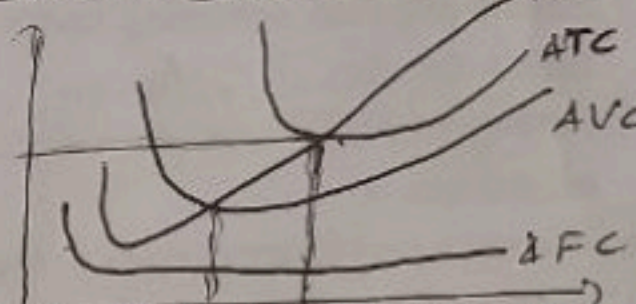
$$Q(Q - 6) = 0$$

$$Q = 0 \quad Q = 6$$

$$MC = 0$$

- 30 = 40 = MR
- P = ?
- A) The consumption level of one commodity can be increased only by reducing the consumption level of the other commodity ~
- B) Marginal rate of substitution between any two commodities (say, X and Y) diminishes
- C) The utility level of a consumer decreases as he/she moves along the same indifference curve
- D) The utility level of a consumer increases as he/she moves along the same indifference curve
- 9) Identify the wrong statement from the following alternatives.
- A) Marginal rate of substitution diminishes as the consumer successively substitutes one commodity for the other commodity ~
- B) A consumer is at equilibrium at the point where his/her budget line is tangent to the highest possible indifference curve
- C) An increase in the income of the consumer makes the slope of a budget line flatter keeping the prices of the commodities constant
- D) Keeping income and price of one good constant, the change in the price of the other good makes the budget line of to rotate on the axis the good whose price has changed

- 10) The output that necessarily maximizes the profit of the firm is at marginal revenue is equal to marginal cost in which



- A) Marginal cost is upward sloping Slope > 0
- B) Marginal cost assumes zero slope
- C) Marginal cost is downward sloping
- D) Marginal revenue is upward sloping

- 11) Which of the following statement is wrong concerning the stages of the short run production?

- A) In the first stage, the size of fixed input is too small to efficiently run the variable input so that this stage of production is inefficient (under utilized)
- B) The second stage is the stage of optimal combination of the variable and fixed factor so that it is the efficient region of production
- C) The third stage is the stage where the fixed input is over utilized, and hence it is inefficient region of production ~
- D) Graphically, the efficient region of production ranges over the point where total product increases at a decreasing rate up to the maximum point of total product curve

- 12) Which of the following is not true about Short run equilibrium of perfect competitive firms:

- A) The profit maximizing level of output is that level of output at which the vertical distance between the TR and TC curves is minimum TR
- B) The profit maximizing level of output is that level of output at which the vertical distance between the TR and TC curves is maximum
- C) The profit maximizing level of output is that level of output at which: P=MC and MC is increasing

- D) B and C

- 13) A price-taking firm implies:

- A) Cannot influence the price of the product it sells.
- B) Talks to rival firms to determine the best price for all of them to charge.

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TVC = 25

A) Average variable cost curve

B) Marginal cost curve

C) Average fixed cost curve (AFC)

D) Average total cost curve

22) A firm's short-run total costs increase from \$45 to \$55 when it increases its production from one unit to two units. Which of the following is true if the total fixed cost is \$30?

A) The average total cost of producing two units is equal to \$47.50

B) Total variable cost is equal to \$25 when two units are produced

C) The marginal cost of producing the first unit is \$10

D) Total variable cost is equal to \$15 when two units are produced

$$MC = \frac{TC_2 - TC_1}{Q_2 - Q_1} = \frac{55 - 45}{2 - 1} = 10$$

$$ATC = \frac{55}{2} = 27.5$$

23) Which of the following statements is CORRECT?

A) The slope of the budget line is equal with the price ratio of the two commodities

B) A combination of two good outside the budget line curve implies affordability

C) The slope of the budget line shows there is no tradeoff between the two goods because the consumer can buy each of them

D) If the consumer's budget increases, the budget line shifts leftward and its slope does change

24) Average variable cost and average total cost get closer together as output increase because?

A) Marginal cost decrease as output increase

B) Diminishing return set in

C) Economies of scale become apparent

D) Average fixed cost fall as produced output increase

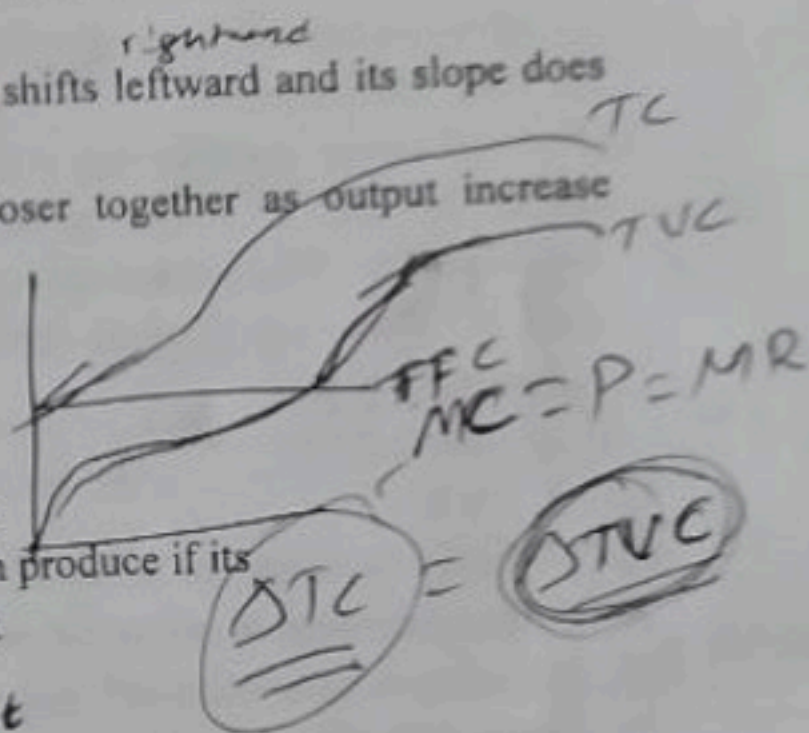
25) A perfectly competitive firm will shut down rather than produce if its

A) Average total cost exceeds its average variable cost

B) Loss is greater than its fixed cost

C) Economic profit is equal to zero

D) Marginal cost curve is starting to rise



26) Which one of the following statement is false?

A) The MC curve passes through the minimum point of both AC and AVC

B) The vertical distance between AC and AFC decrease continuously as output increases.

C) The ATC curve reaches its minimum point at higher output than AVC reaches its minimum point

D) The change in total cost is equal to the change in total variable cost in short run

27) Which of the following would create a natural monopoly?

A) Requirement of a government license before the firm can sell the good or service

B) Technology enabling a single firm to produce at a lower average cost than other firms

C) An exclusive right granted to supply a good or service

D) Ownership of all the available units of a necessary input

28) The law of diminishing returns states that:

A) As a firm uses more of a variable resource, given the quantity of fixed resources, the average product of the firm will increase

B) As a firm uses more of a variable resource, given the quantity of fixed resources, marginal product of the variable factor will eventually decrease

C) In the short run, the average total costs of the firm will eventually diminish

$$\frac{P_x X + P_y Y}{P_x} = \frac{M}{P_x}$$

$$\frac{P_x X + P_y Y}{P_y} = \frac{M}{P_y} \Rightarrow Y = \frac{M}{P_y} - \frac{P_x}{P_y} X$$

$$P_x = 5$$

$$P_y = 5$$

10 =

C) Sets the product's price to whatever level the owner decides upon.

D) Asks the government to set the price of its product.

14) Bikila's marginal utility from consumption of good X is 200 and his marginal utility from the consumption of good Y is 100. Bikila spends his entire budget. The price of good X is \$5 per unit and the price of good Y is \$5 per unit. To maximize his utility, Bikila should

A) Buy less Y and more X.

B) Buy less Y and fewer X.

C) Buy more Y and fewer X.

D) Buy more Y and more X.

15) Which of the following statement is incorrect about average variable cost, average fixed cost average total cost, and marginal cost?

A) Average variable cost reaches minimum before the minimum of average total cost. ~

B) Always average variable cost and average total cost have positive relationship

C) Marginal cost passes through the minimum point of all per unit costs?

D) Both average and marginal costs have U-shape due to the law of variable proportion ~

16) All of the following statements are incorrect about imperfect market structure except: True

A) Monopolistic competitive market structure is characterized by differentiated products without close substitutes

B) Monopolistic competitive market structure is characterized by differentiated products with close substitutes

C) Firm's decision operating under oligopoly market structure is independent of other firm's decision x Collusion

D) Pure monopoly market structure is characterized by heterogeneous products with close substitutes x

17) Indifference curves are convex to the origin because they are based on

A) Increasing marginal rate of substitution

B) Diminishing marginal rate of substitution

C) Constant rate of marginal rate of substitution

D) Zero marginal rate of substitution

18) The common characteristics of all market structures is:

A) Large number of buyers ~

C) Large number of sellers

B) Entry and exit condition

D) Type of products

19) Which of the following statement is false in general about market structures in the short run?

A) The supply curve of a perfectly competitive firm is the part of marginal cost above average cost curve

B) The firm decides to shut down its operation when its total revenue fails to cover the variable cost ~

C) The equilibrium condition of monopoly firm is obtained by equating the marginal revenue and marginal cost

D) A perfectly competitive firm equates its marginal cost with market price to maximize its profit ~

20) Macroeconomic problem does not include

A) High Inflation.

B) High unemployment.

C) Slow economic growth.

D) High population growth.

21) A cost curve that can never be "U" shaped is the:



$$MR = MC$$

$$-3Q^2 + 12Q + 40 = 40$$

$$-3Q^2 + 12Q = 0 \Rightarrow -3(Q^2 - 4Q) = 0$$

$$Q^2 - 4Q = 0 \Rightarrow Q(Q - 4) = 0$$

$$Q = 0$$

and

$$Q = 4$$

Slope of MC ~~at Q=0~~

$$\frac{d(MC)}{dQ} = -6Q + 12$$

at $Q=0$, $-6(0) + 12 = 12 > 0$ --- True

at $Q=4$, $-6(4) + 12 = -12 < 0$ --- False

Therefore, at $Q=0$ ~~the~~ will be the maximizing Profit level

(b)

$$\pi = TR - TC$$

$$\pi = 40Q - (-Q^3 + 6Q^2 + 40Q + 20)$$

$$\pi = 40Q + Q^3 - 6Q^2 - 40Q - 20$$

$$\pi = Q^3 - 6Q^2 - 20$$

$$\pi \text{ at } Q=0 = \underline{-20}$$

52 loss

Therefore, it is a loss

(c) The shut down Price is: $AVC = P$

$$AVC = \frac{TVC}{Q} = \frac{-Q^3 + 6Q^2 + 40Q}{Q} = -Q^2 + 6Q + 40$$

$$-Q^2 + 6Q + 40 = 40 \Rightarrow -Q^2 + 6Q = 0$$

$$-(Q^2 - 6Q) = 0 \Rightarrow Q(Q - 6) = 0$$

$$Q = 0$$

$$Q = 6$$

Slope of AVC $-2Q + 6$

at $Q=0$, $-2(0) + 6 = 6 > 0$

at $Q=6$, $-2(6) + 6 = -6 < 0$

Therefore $Q=0$

is the Shut-down point

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